

NGO Volunteer Impact Hub

A full-stack ServiceNow-powered volunteer management platform

Project Overview | 2025

1. Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + Vite	Single-page application UI
Backend / Database	ServiceNow PDI	Tables, Business Rules, REST API
Production Hosting	Vercel	Static site + serverless proxy function
API Layer	ServiceNow Table API	CRUD operations on all 7 tables
Dev Proxy	Vite Dev Server	Forwards /api/now/* to ServiceNow in development
Prod Proxy	Vercel Serverless (api/proxy.js)	Injects Basic Auth, forwards to ServiceNow

2. System Architecture

Browser → React App (Vercel CDN) | Sends /api/now/* requests (no credentials in browser) ▼ Vercel Serverless Function (api/proxy.js) | Injects: Authorization: Basic <base64 admin:password> ▼ ServiceNow PDI (dev286774.service-now.com) | Business Rules fire automatically on INSERT / UPDATE / DELETE ▼ 7 Custom Tables + sys_user + 5 Business Rules

3. ServiceNow Tables (7 Tables)

Table Name	Purpose
sys_user	Volunteer & Admin accounts (built-in SN table, extended with 5 custom fields)

x_2048396_ngo_vo_1_ngo_projects	NGO project records with name, location, dates, skills, status
x_2048396_ngo_vo_1_volunteer_assignments	Links volunteers to projects — tracks hours and completion status
x_2048396_ngo_vo_1_impact_report	Impact reports with volunteers involved, hours, beneficiaries, outcome summary
x_2048396_ngo_vo_1_events	Volunteer events with date, time, location, max participants, registered count
x_2048396_ngo_vo_1_event_registrations	Links volunteers to events — triggers Business Rules on insert/delete
x_2048396_ngo_vo_1_notifications	In-app notifications (approval, assignment, event) per volunteer

Custom Fields Added to sys_user

Field	Values
x_2048396_ngo_vo_1_u_is_volunteer	true / false — marks a user as a volunteer
x_2048396_ngo_vo_1_u_approval_status	pending / approved / rejected
x_2048396_ngo_vo_1_u_skills	comma-separated skills (Teaching, First Aid, IT Support...)
x_2048396_ngo_vo_1_u_availability	weekdays / weekends / flexible / mornings / evenings
x_2048396_ngo_vo_1_u_address	volunteer address (City, State)

4. ServiceNow Business Rules (Server-Side Automation)

Rule	Table	Trigger	Action
BR-1	event_registrations	After INSERT	Increments <code>registered_count</code> on the parent event record
BR-2	event_registrations	Before DELETE	Decrements <code>registered_count</code> on the parent event record
BR-3	sys_user	After UPDATE when <code>approval_status</code> → approved	Creates an approval notification for the volunteer
BR-4	volunteer_assignments	After INSERT	Creates an assignment notification for the assigned volunteer
BR-5	event_registrations	After INSERT	Creates an event registration notification for the volunteer

5. Login & Role Separation

 **Admin Role**

Logs in via the **Admin** tab. If a volunteer account tries the Admin tab — blocked with an error message.

Can access:

Dashboard · Volunteers · Projects · Assignments · Events · Impact Reports · Notifications

 **Volunteer Role**

Logs in via the **Volunteer** tab. Must be `active=true` and `approval_status=approved` in ServiceNow.

Can access:

Volunteer Dashboard · Notifications

New Volunteer Sign-Up Flow

1. Volunteer signs up on login page

→ `sys_user` record created: `active=false`, `approval_status=pending`

2. Admin sees them in "Pending Approval" tab → clicks Approve

→ `active=true`, `approval_status=approved`

→ BR-3 fires → notification created: "Your registration has been approved"

3. Volunteer can now log in and access the Volunteer Dashboard

6. Admin Portal — Pages & Features

Page	Features
Dashboard	Live stats (volunteers, projects, assignments, events) · ServiceNow connection status · Test SN Connection button pings all 7 tables

Volunteers	Active tab: searchable + paginated table, View Stats modal · Pending tab: Approve / Reject buttons · Register Volunteer (admin-approved instantly)
Projects	Card grid with search & status filter · Full CRUD (create, edit, delete) · Skill tags on cards · Quick-add skill buttons in form
Assignments	Paginated table · Full CRUD · Skill-match suggestion (matching volunteers highlighted) · Hours progress bar per row
Events	Full CRUD · Registered count auto-maintained by BR-1 / BR-2 · Capacity tracking per event
Impact Reports	Aggregate stats strip · Tabbed bar chart (Beneficiaries / Hours / Volunteers per project) · Export report as .txt · Delete
Notifications	Shows all notifications · Mark as read / Mark all as read

7. Volunteer Portal — Sections & Features

Section	Features
Stats Row	Total Volunteers · My Assignments · My Projects · Registered Events
Volunteer Registration Form	Always-visible form: Full Name, Phone, Skills (with quick-add buttons), Availability, Address · Saves directly, no approval message shown
My Profile	View / Edit toggle for all profile fields · Saves to ServiceNow
My Assignments	Lists assignments with inline hours logging — input + Log Hours button syncs hours to ServiceNow
My Projects	Projects the volunteer is currently assigned to
Available Events	Events open for registration · Spots-left counter · Register button triggers BR-1 (count++) and BR-5 (notification)
Notifications	Approval, assignment, and event notifications · Mark as read

8. Key Data Flows

Admin Creates an Assignment

```
Admin clicks "Create Assignment"
→ POST /x_2048396_ngo_vo_1_volunteer_assignments
→ BR-4 fires in ServiceNow
→ notification record created for the volunteer (type = assignment)
→ Volunteer sees: "You have been assigned to project: [name]"
```

Volunteer Registers for an Event

```
Volunteer clicks "Register"
→ POST /x_2048396_ngo_vo_1_event_registrations
→ BR-1 fires → event.registered_count + 1
→ BR-5 fires → notification created (type = event)
→ Spots-left counter decrements on next page load
```

Volunteer Unregisters from an Event

```
DELETE /x_2048396_ngo_vo_1_event_registrations/{id}
→ BR-2 fires → event.registered_count - 1
→ Spot becomes available again for other volunteers
```

Admin Approves a Volunteer

Admin clicks "Approve" on Pending tab

→ PATCH /sys_user/{id} { active: true, approval_status: approved }

→ BR-3 fires → notification created: "Your volunteer registration has been approved. Welcome to the team!"

→ Volunteer can now log in with their credentials

9. Deployment

Environment	URL	API Proxy
Local Development	http://localhost:5173	Vite dev proxy in vite.config.js
Production	https://ngo-impact-hub.vercel.app	Vercel serverless function (api/proxy.js)

Both environments inject the ServiceNow Basic Auth header server-side. Credentials are never exposed in the browser JavaScript bundle.

https://ngo-impact-hub.vercel.app

10. Credentials

🔑 Admin Account

Username: admin

Password: p/rLsoR41AV^

Tab: Admin

👤 Volunteer Account

Username: any approved volunteer email

Password: their registered password

Tab: Volunteer

11. Key Project Files

File / Folder	Purpose
src/api/servicenow.js	All ServiceNow REST API calls — auth, CRUD for all 7 tables, demo fallback data
src/App.jsx	Login logic, role validation, portal routing (admin vs volunteer)
src/components/Sidebar.jsx	Navigation sidebar — filters nav items by adminOnly flag
src/pages/VolunteersPage.jsx	Admin: volunteer management, approval, stats modal, pagination
src/pages/ProjectsPage.jsx	Admin: project CRUD with search and status filters

<code>src/pages/AssignmentsPage.jsx</code>	Admin: assignment CRUD with skill-match suggestions, pagination
<code>src/pages/EventsPage.jsx</code>	Admin: event CRUD
<code>src/pages/ReportsPage.jsx</code>	Admin: impact reports with tabbed bar charts
<code>src/pages/VolunteerDashboard.jsx</code>	Volunteer: registration form, assignments, events, hours logging
<code>api/proxy.js</code>	Vercel serverless function — proxies <code>/api/now/*</code> to ServiceNow with auth
<code>vercel.json</code>	Vercel config — rewrite rules for proxy and SPA routing
<code>vite.config.js</code>	Vite dev proxy config — forwards <code>/api/now/*</code> to ServiceNow in development